

OSHIMA, Japan

WMO: 476750

Lat: 34.7494N

Lon: 139.3631E

Elev: 248

StdP: 14.56

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	34.3	36.5	18.1	13.8	41.5	21.4	16.2	42.0	31.8	46.7	29.8	47.1	9.0	80	0.511

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.4	86.8	79.1	85.4	78.4	83.9	77.7	80.5	84.4	79.6	83.4	78.7	82.5	11.2	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.5	154.3	83.1	78.5	149.3	82.3	77.5	144.5	81.4	44.4	84.5	43.5	83.5	42.5	82.8	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.8	24.8	21.7	DB	29.4	89.7	1.7	1.2	28.1	90.6	27.1	91.3	26.1	92.0	24.9	92.9	(4)
(5)			WB	26.2	81.9	2.0	1.3	24.7	82.8	23.5	83.6	22.3	84.3	20.8	85.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	61.8	45.2	45.8	50.6	57.9	64.9	70.1	77.1	79.6	74.5	66.5	58.0	49.8
	DBStd	12.48	4.02	4.71	5.25	5.01	3.66	3.52	3.88	3.15	4.49	4.37	4.72	4.78
	HDD50	420	159	135	58	5	0	0	0	0	0	0	1	62
	HDD65	2584	614	539	446	217	45	4	0	0	1	33	214	471
	CDD50	4711	9	17	76	242	463	603	839	917	736	511	241	57
	CDD65	1399	0	0	0	3	44	156	374	452	287	79	4	0
	CDH74	8722	0	0	0	0	33	320	2613	3975	1620	161	0	0
	CDH80	1695	0	0	0	0	0	15	487	940	248	5	0	0
Wind	WSAvg	11.1	12.2	11.5	12.2	11.6	10.5	9.6	10.1	9.7	10.4	11.2	11.4	12.2
Precipitation	PrecAvg	100.8	4.9	5.8	8.4	8.8	9.3	11.5	8.8	7.7	11.9	11.5	7.8	4.5
	PrecMax	128.9	12.0	10.7	16.5	15.9	18.7	20.6	22.3	20.1	19.6	33.2	21.1	8.5
	PrecMin	67.3	0.4	1.1	3.0	3.7	3.6	5.2	0.2	0.2	3.5	2.2	2.3	0.4
	PrecStd	15.1	2.9	2.9	3.1	3.2	3.7	4.6	5.5	5.2	4.5	6.6	4.5	2.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.0	63.3	66.8	71.9	77.2	82.1	88.2	89.0	86.6	80.6	72.4	67.3
		MCWB	56.0	58.0	58.9	63.1	67.3	74.8	78.9	80.3	78.4	72.6	65.5	63.1
	2%	DB	57.8	59.8	64.1	69.6	74.6	79.0	86.2	87.3	84.6	77.8	69.9	63.2
		MCWB	51.9	54.8	57.8	62.9	66.6	72.5	78.6	79.5	77.5	71.3	63.6	57.5
	5%	DB	55.1	57.0	62.1	67.8	73.0	77.2	84.6	86.1	82.9	75.6	68.1	60.5
		MCWB	48.2	50.9	56.7	61.7	65.8	71.4	77.9	79.0	76.6	70.1	62.2	53.7
	10%	DB	52.6	54.4	60.2	66.1	71.5	75.7	82.8	84.7	81.1	73.6	66.0	57.9
		MCWB	45.6	47.9	54.5	59.9	65.2	70.7	77.2	78.4	75.8	68.0	60.1	51.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.8	60.0	62.2	68.2	71.9	76.6	81.1	82.2	80.1	75.4	68.4	64.7
		MCDB	59.8	61.9	63.7	69.2	73.3	79.0	85.4	86.4	83.9	78.0	70.3	66.1
	2%	WB	53.1	55.9	59.9	65.3	69.7	74.8	79.8	81.0	78.9	73.3	65.8	58.8
		MCDB	56.9	58.6	62.8	67.7	71.9	76.9	83.9	84.9	82.6	76.3	68.5	62.1
	5%	WB	49.2	52.2	57.7	63.3	68.2	73.5	78.8	80.2	77.9	71.1	63.4	54.9
		MCDB	53.8	56.4	61.3	66.4	71.0	75.8	82.8	83.9	81.4	74.8	66.7	59.1
	10%	WB	46.4	48.7	55.2	61.2	66.6	72.1	77.9	79.4	76.9	68.9	61.3	52.2
		MCDB	51.7	53.3	59.4	64.9	70.0	74.7	81.6	83.1	80.2	72.7	65.0	56.5
Mean Daily Temperature Range	5% DB	MDBR	11.3	12.0	12.2	12.2	11.4	9.3	8.8	9.4	9.4	9.4	10.4	11.1
		MCDBR	13.8	14.3	13.8	13.5	13.5	11.3	10.6	10.7	11.3	11.6	12.6	13.5
		MCWBR	10.5	11.7	11.1	9.2	8.5	6.6	5.4	5.5	6.3	7.8	9.2	10.2
	5% WB	MCDBR	13.0	13.4	12.6	10.9	10.7	8.4	9.5	9.7	10.2	10.5	11.3	12.6
		MCWBR	11.5	12.8	12.0	9.4	8.4	5.7	5.1	5.5	6.3	8.3	9.9	11.1
Clear-Sky Solar Irradiance	taub	0.382	0.430	0.488	0.532	0.544	0.560	0.555	0.540	0.493	0.478	0.425	0.382	
	taud	2.255	2.119	1.976	1.912	1.947	1.988	2.048	2.082	2.183	2.131	2.202	2.263	
	Ebn at Noon	254	253	248	244	242	238	238	239	245	235	236	245	
	Edh at Noon	34	43	54	61	59	57	53	51	44	43	36	32	
All-Sky Solar Radiation	RadAvg	836	1007	1279	1499	1634	1427	1650	1669	1296	975	792	749	
	RadStd	65	109	85	162	173	142	258	135	111	102	69	62	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	+1.44	N/A	+0.50	N/A	N/A	N/A	N/A	N/A
Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page